ESR1 ligand-binding domain: structural diversity of the crystal structures

Results since the October report: requests of 28 October, 4 November and 11 November

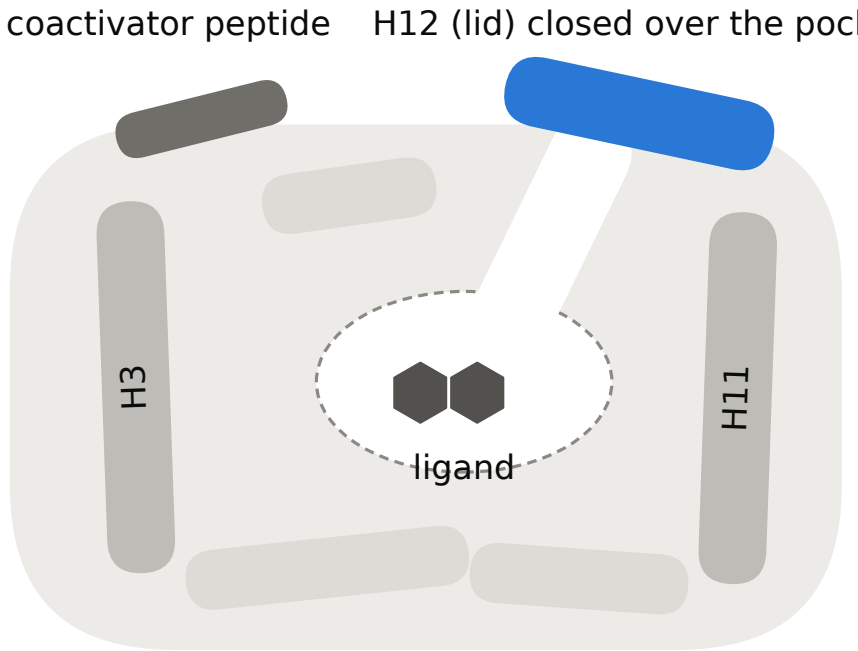
13 November 2026

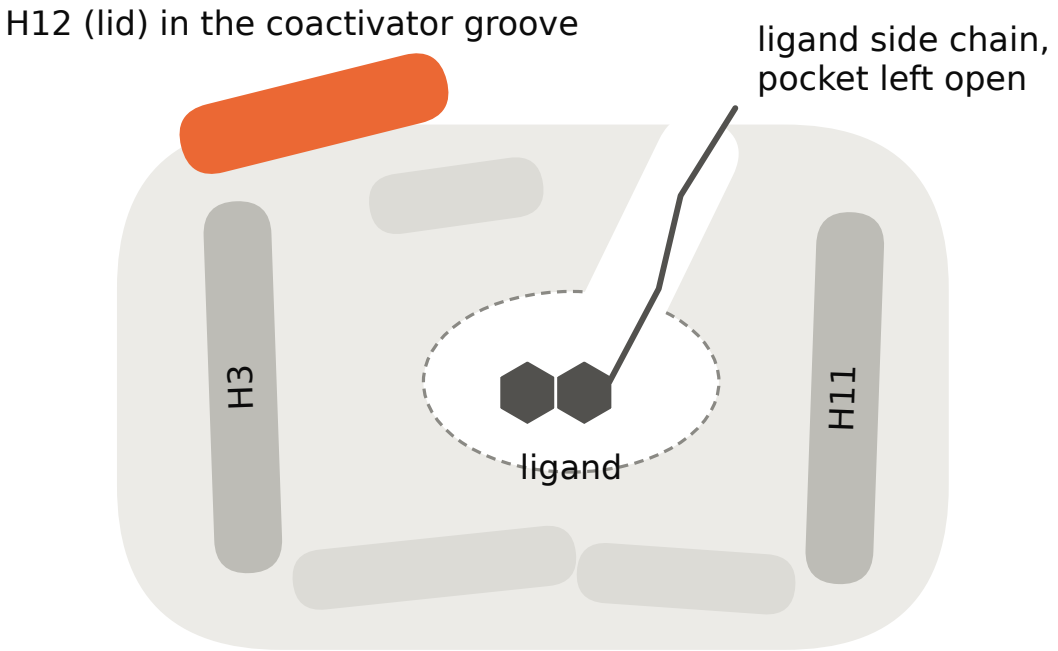
788 chains of oestrogen receptor α from 351 PDB entries

What was asked at the last three meetings, and where each answer is

Meeting	Request	Slide
28 October	Does lid completeness follow the crystal or the biology?	**4, 15**
	Helix ranges from consensus secondary structure	**4**
	Compare incomplete helices by their midpoints	**5**
	Colour the outliers in the clustering	**8, 10**
4 November	Action classes against structure	**6**
	Compare affinities measured in different assays	**7**
	Pocket filling relative to the E2 pocket	no group difference (q 0.14)
	Does an antagonist destabilise H12?	open: class follows lid (6)
11 November	Outliers in lid-to-H11 distance: group 4, group 7, grey	**8, 9**
	Sammon map, self-organising map, UMAP of PCA	**10**
	Hide H12 and cluster again	**11**
	Phi/psi, helix centres of mass, no superposition	**11, 12, 13**
	Do crystallisation additives stabilise the lid?	**14, 15**
	Biological variability or a real effect?	rule on slide 3

Two lid positions split the chains into the two groups used on every slide: group 7 and group 4





Group 7: lid closed over the pocket

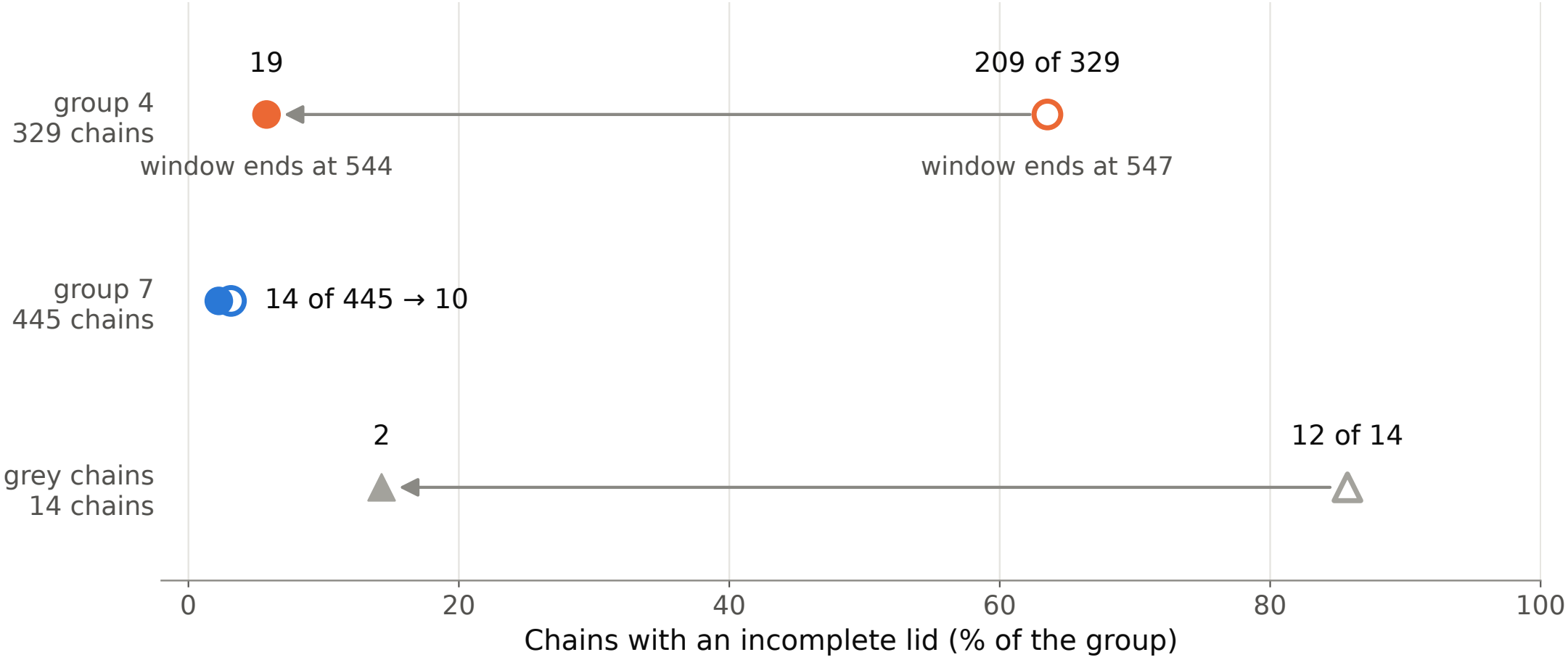
445 chains · lid 12.5 Å from H11 · peptide bound in 414

Group 4: lid in the coactivator groove

329 chains · lid 22.3 Å from H11 · peptide bound in 4

Groups	788 chains from 351 structures, clustered by C-alpha RMSD at 1.5 Å; 14 grey chains form 5 small groups
Rule	a difference counts only if it is larger than the spread within a group and between copies of one crystal
Example	lid-to-H11 distance: copies of one crystal differ by up to 0.91 Å (group 7) and 1.08 Å (group 4)

Ending the lid window at residue 544 instead of 547 makes an incomplete lid rare: in group 4 it drops from 209 to 19 chains

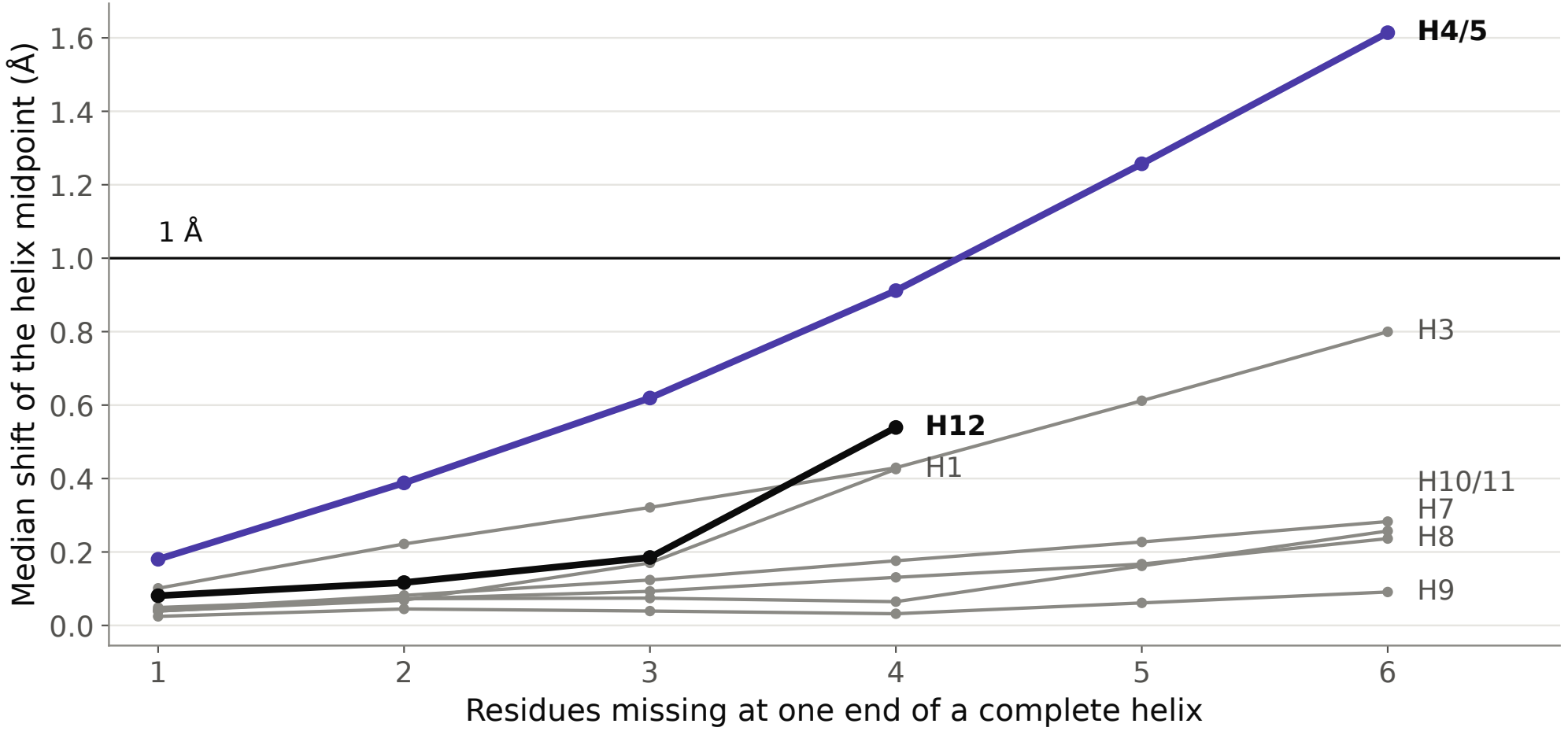


Shows chains whose lid (H12) is partly or not modelled, for the window 538–547 and for 538–544

Means 190 of the 209 incomplete group 4 lids lack only residues 545–547: the lid is there, its last turn is not

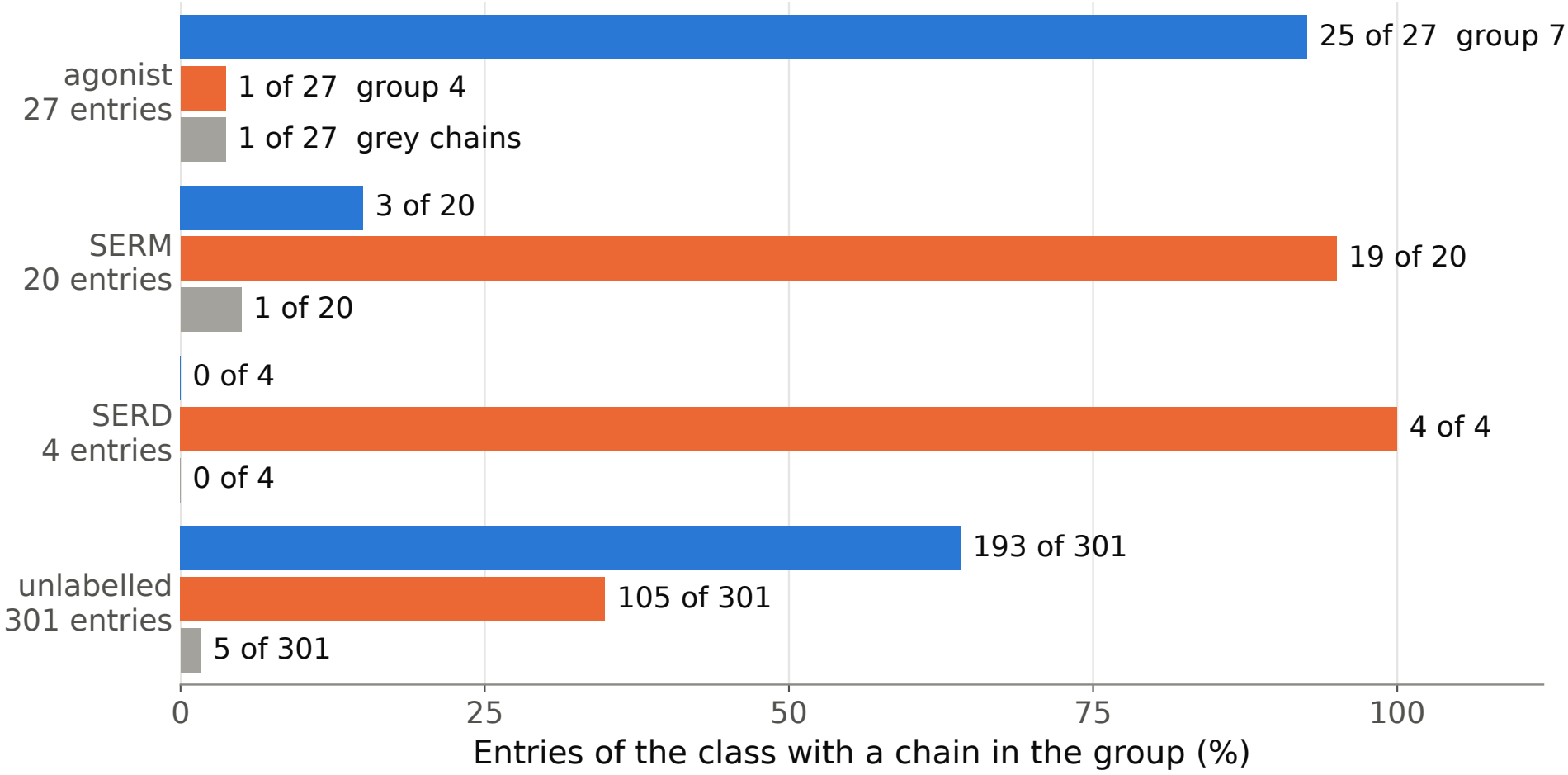
Caveat the shorter window comes from consensus secondary structure and is not yet agreed

A helix midpoint moves 0.91 Å or less (median) when 4 end residues are missing, so helices with missing ends can still be compared



Shows	complete helices cut short by 1–6 residues; how far their midpoint moves
Means	distances between helix midpoints reproduce the two lid groups (adjusted Rand index 0.954)
Caveat	H4/5 crosses 1 Å at 5 missing residues; the agreement mostly restates the H12 split

Action classes almost coincide with the lid groups: agonists 25 of 27 entries in group 7, SERMs 19 of 20 in group 4

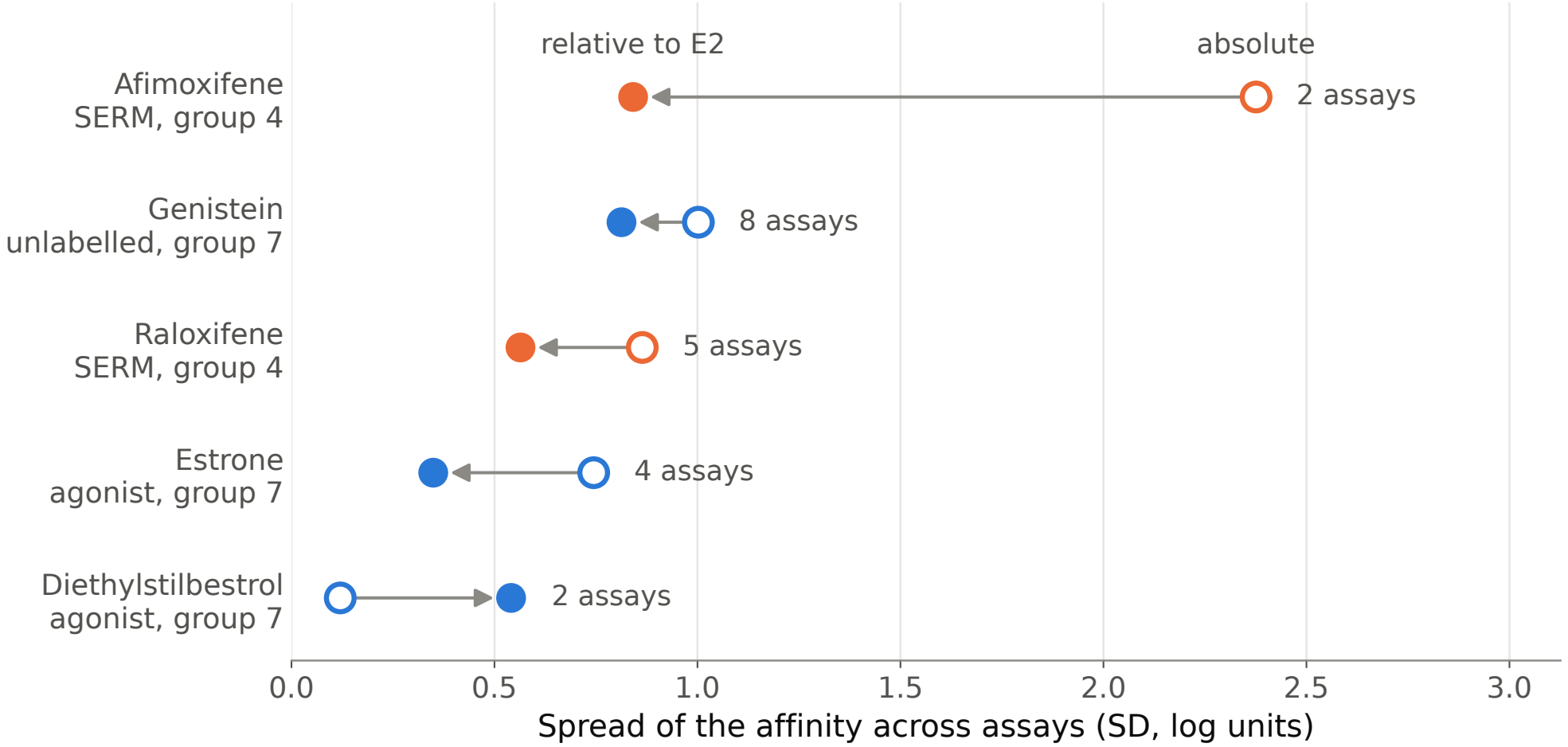


Shows for each action class, the share of its crystal structures that hold a chain in each lid group

Means class cannot yet be tested apart from lid position, nor its link to H12 stability or the outliers

Caveat uncurated ChEMBL labels: only 51 of 351 entries carry a class

Referencing each affinity to E2 in the same assay narrows the spread between assays for 4 of 5 ligands (median SD 0.86 → 0.56)

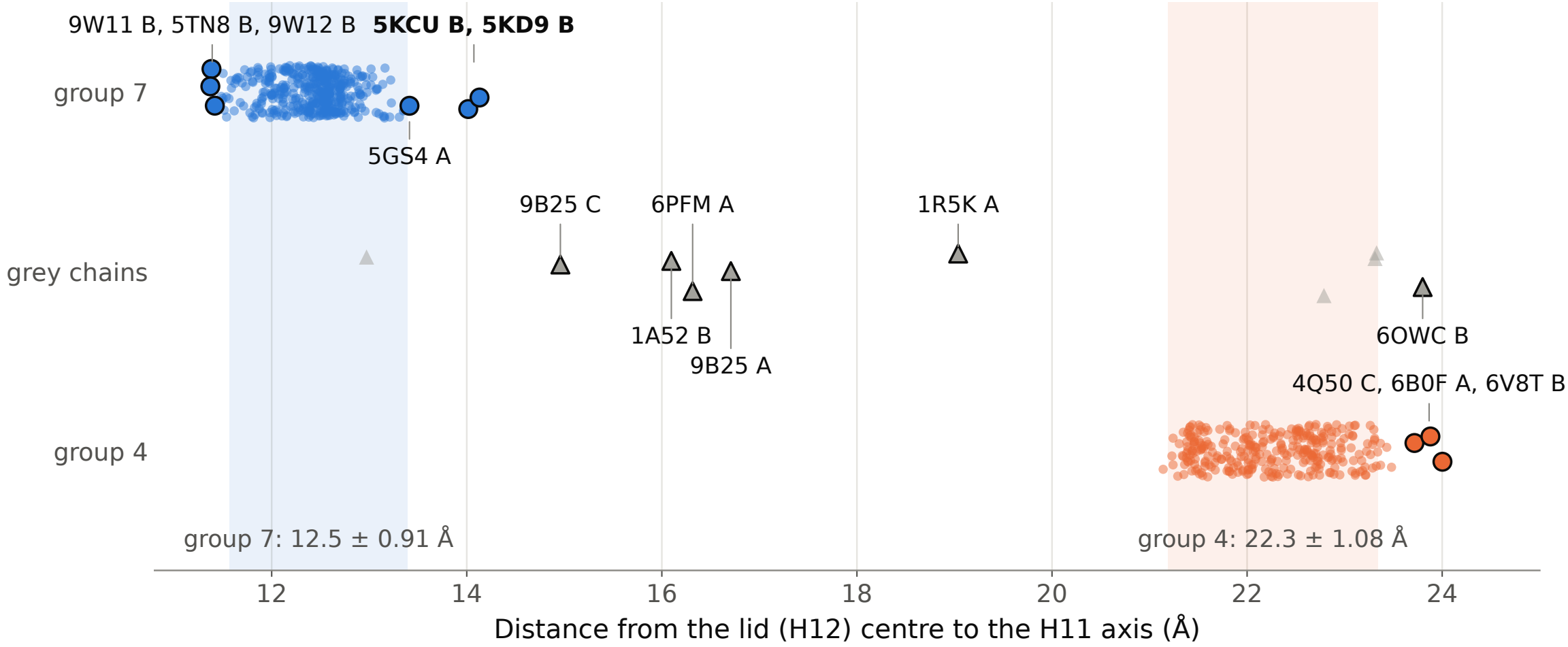


Shows open: absolute affinities from different assays; filled: the same affinities minus E2 in each assay

Means affinities from different systems become more comparable when each is referenced to E2

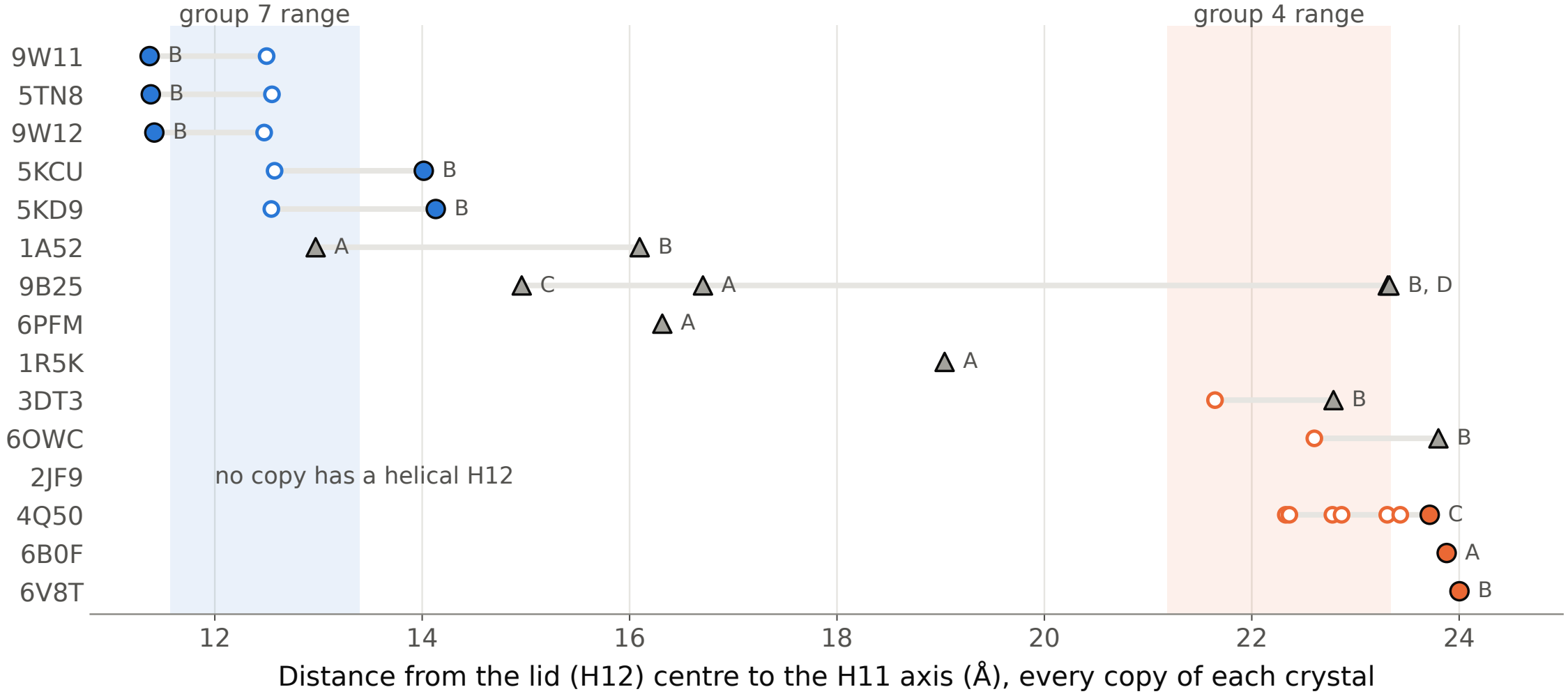
Caveat only 5 ligands have two or more assays that also measured E2

Two group 7 chains, 5KCU B and 5KD9 B, sit furthest out; group 4 has only 3 mild extremes; 5 grey chains hold the lid in between



Shows	every chain; shaded: group median ± the largest normal difference between copies of one crystal
Means	15 chains lie beyond ordinary variation (ringed): these are the structures to inspect one by one
Caveat	40 chains without a helical H12 have no distance

An unusual lid usually belongs to one copy, not to the whole crystal: in 7 of 15 crystals the other copies are ordinary



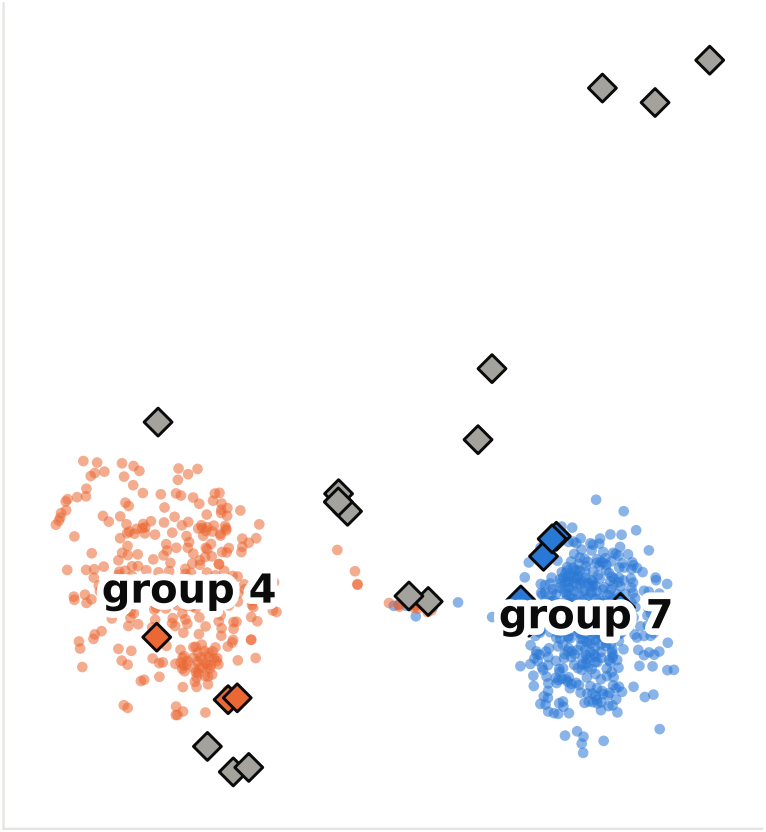
Shows crystals holding an examined chain (filled, letter = copy) next to their other copies (open)

Means what copies share (ligand, drop) does not fix the lid: 9B25 spans 15.0–23.3 Å in one crystal

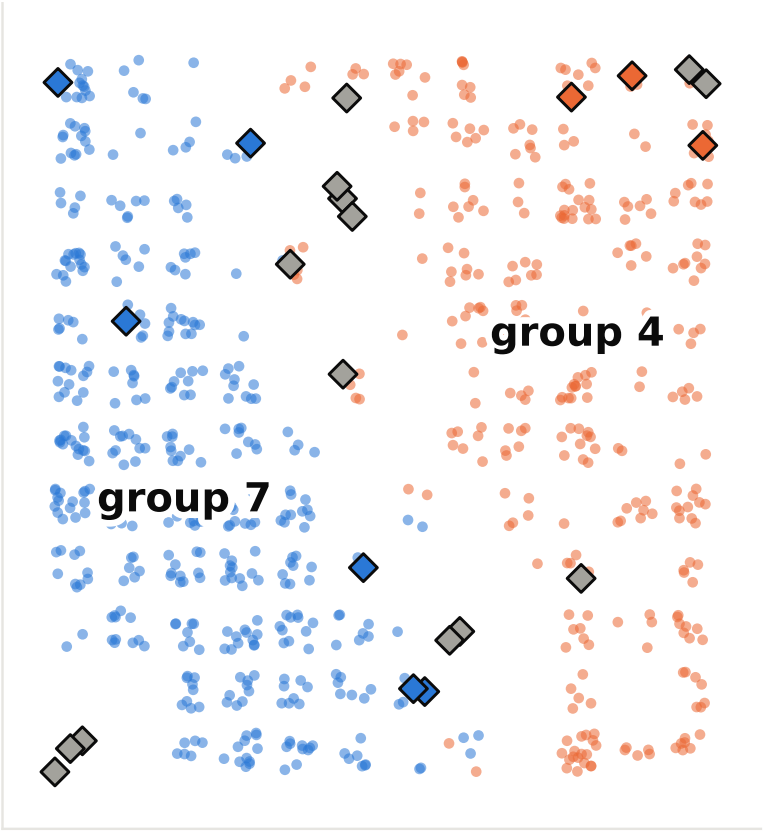
Caveat copies of one crystal sit in different lattice contacts (slide 15)

All three maps show the same two groups; no smaller island stands out more than copies of one crystal differ (0.60 vs 1.26 Å)

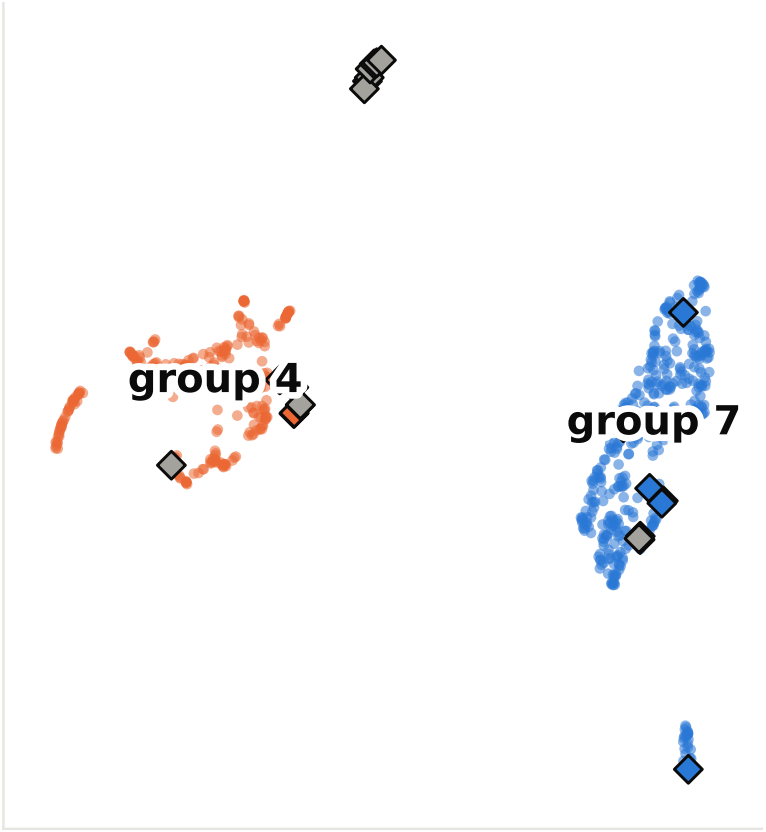
Sammon map
group separation 0.76 (silhouette)



Self-organising map (12 × 12)
group separation 0.45 (silhouette)



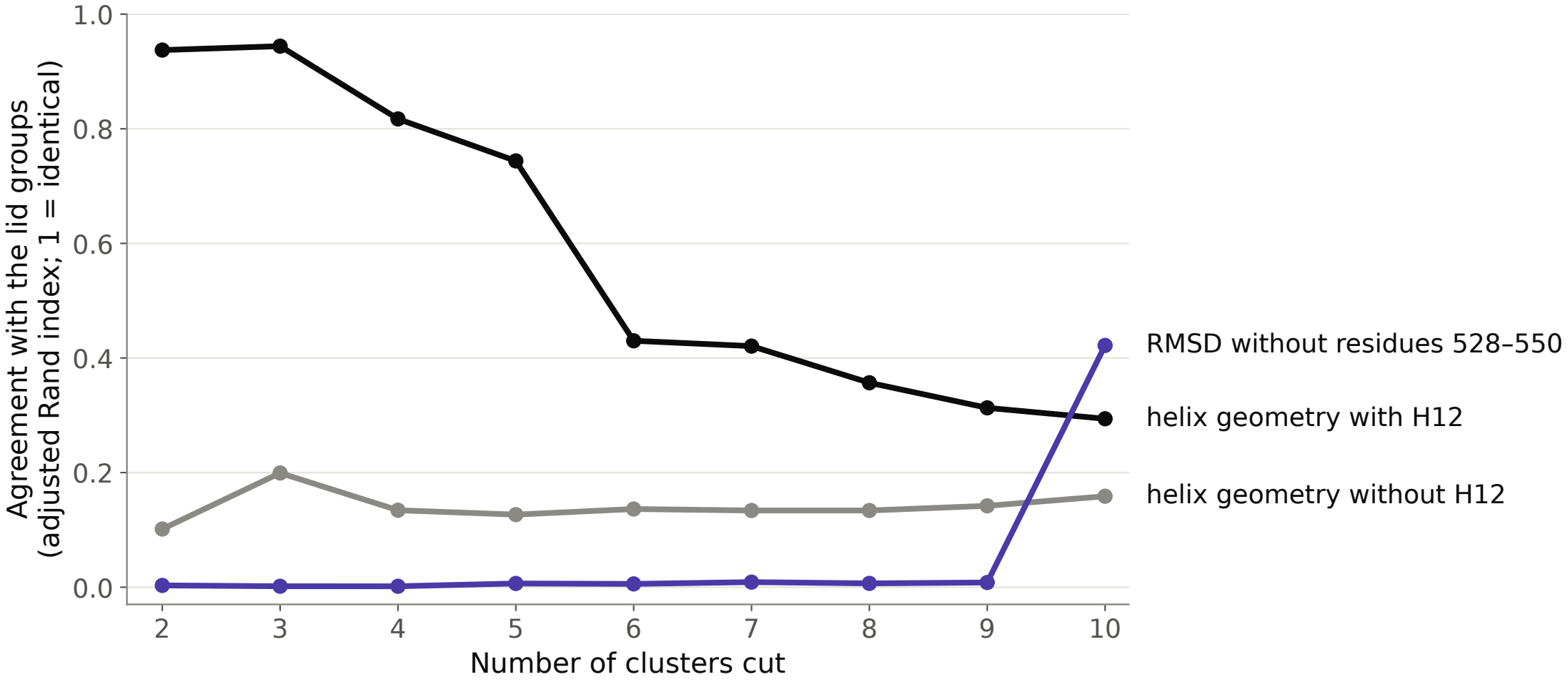
UMAP of 6 PCA scores
group separation 0.76 (silhouette)



Shows each chain placed by its RMSD to all others; diamonds: chains examined as lid outliers or grey

Means the Sammon map keeps RMSD distances best (rank correlation 0.92); the smaller islands follow crystal form

Hide H12 and the groups disappear: helix geometry with H12 recovers them (agreement 0.94), without H12 none exceeds 0.42

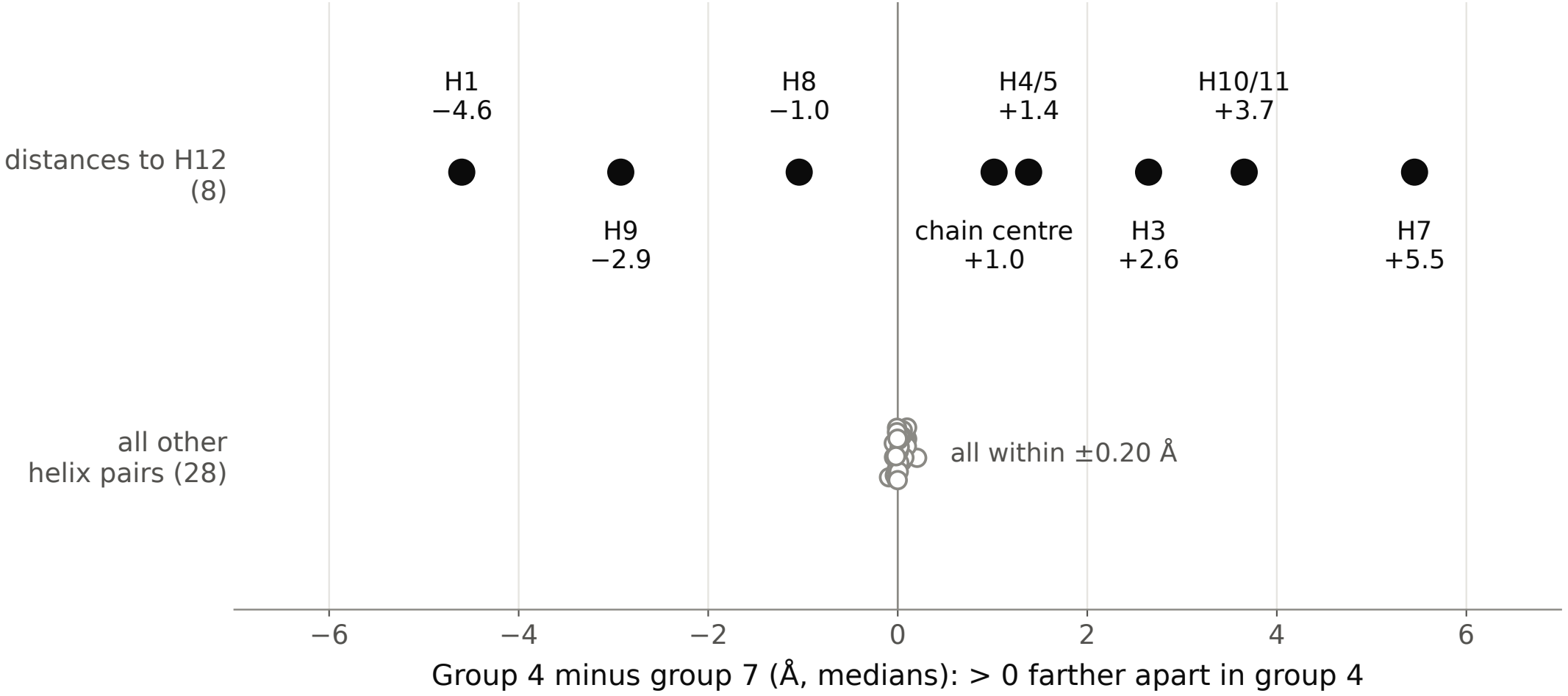


Shows how well each clustering finds group 7 and group 4, for 2 to 10 clusters

Means without the lid the chains form one ensemble: no split exceeds copy-to-copy differences (0.71 vs 1.12 Å)

Caveat the 10-cluster RMSD split partly follows the lid groups (0.42) but is smaller than copy differences

Without superposition only distances to H12 change: up to 5.5 Å, while all 28 other helix pairs move 0.20 Å or less

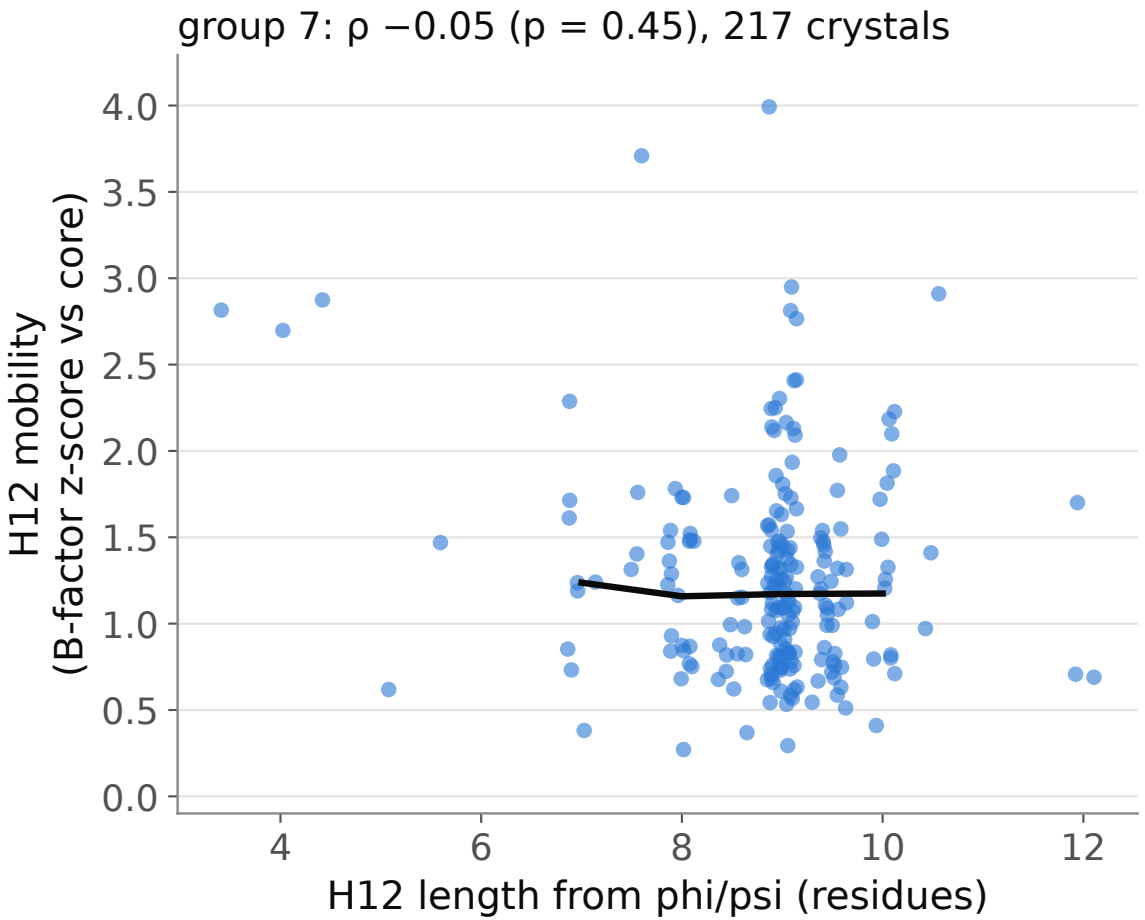


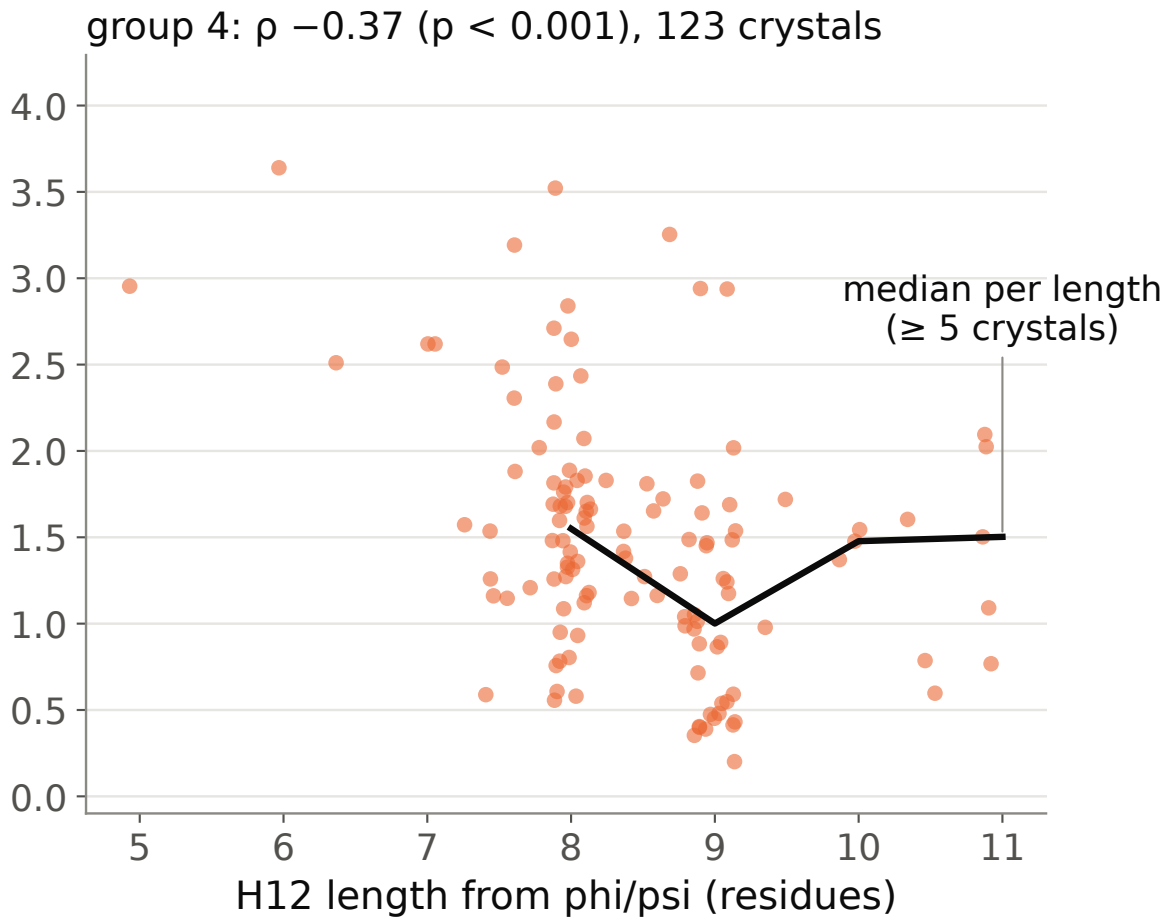
Shows distances between helix centres of mass from the deposited coordinates; filled = beyond ordinary variation

Means 8 of 36 distances change, all involving H12: the rest of the domain keeps its shape

Caveat a helix centre needs at least 70% of the helix modelled

In group 4 a longer H12 is a less mobile H12 (ρ −0.37, 123 crystals); in group 7 there is no link (ρ −0.05)



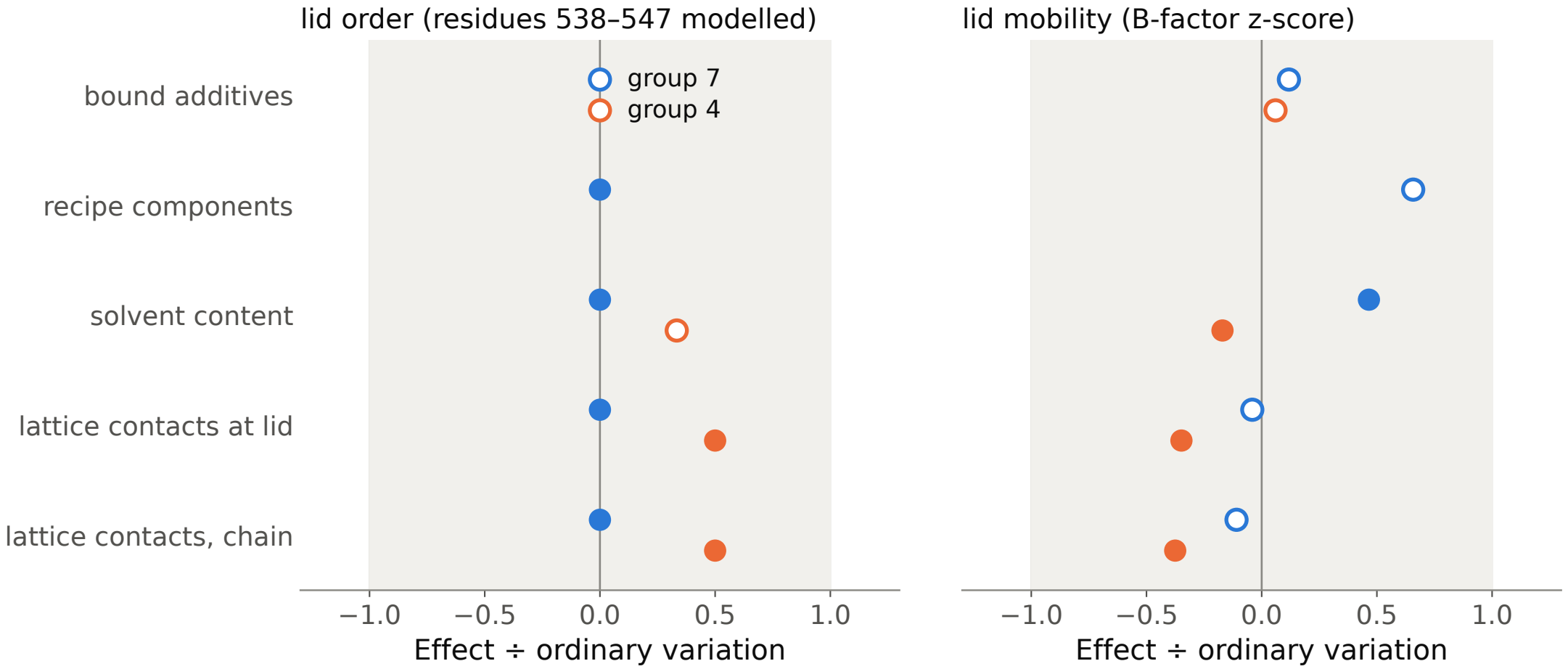


Shows one point per crystal: helical length of H12 against how mobile it is

Means in the groove, a partly unwound lid is also a looser lid; in group 7 H12 is almost always complete

Caveat B-factors also depend on refinement (TLS) and resolution

Additives, recipe, solvent content and lattice contacts never move lid order or mobility beyond ordinary variation (0 of 22 tests)

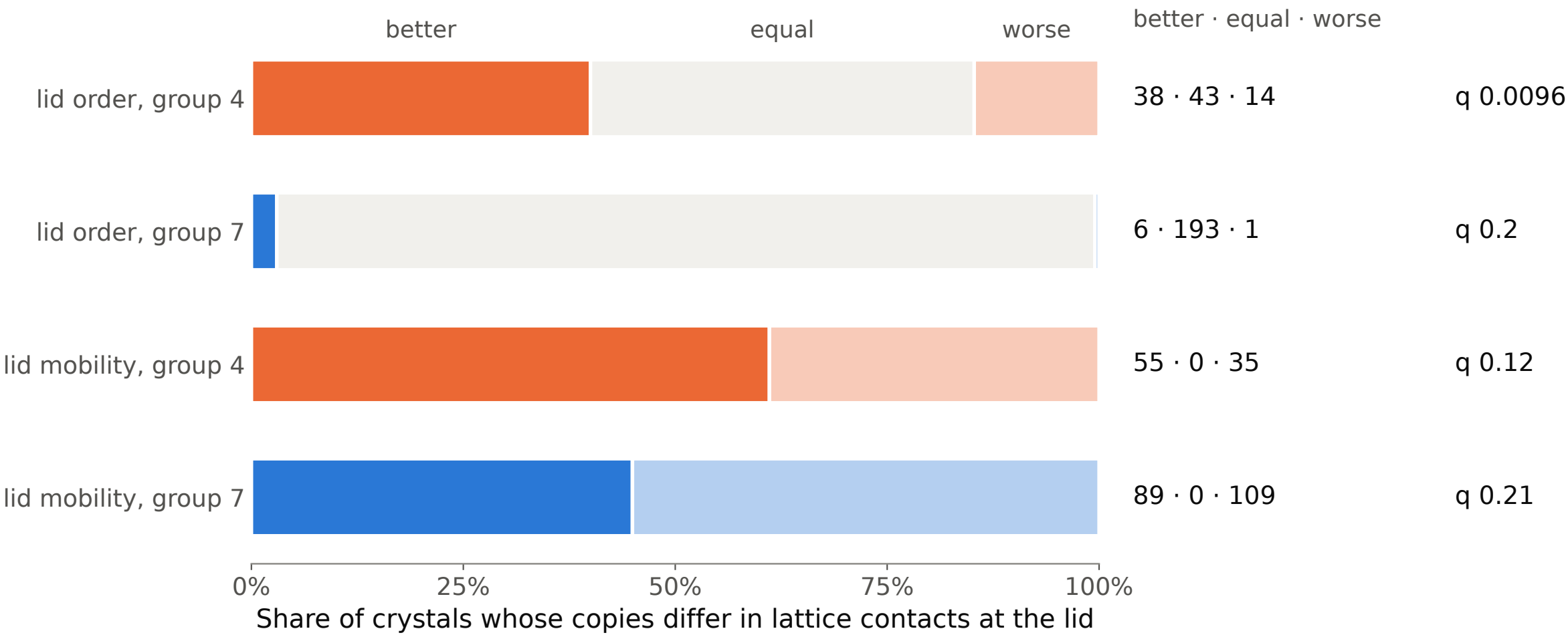


Shows shaded ±1 = ordinary variation; filled = detectable (q < 0.05) yet smaller than ordinary variation

Means more components in the drop do not make the lid more ordered or less mobile (largest 0.66 of ordinary)

Caveat bound additives are rare: median 0 per chain, 50 of 351 entries have any

Within one crystal, the group 4 copy with more lattice contacts at the lid has more of the lid modelled: 38 crystals against 14



Shows copies share the drop, additives and solvent: does the copy with more contacts have the better lid?

Means the crystal lattice, not the recipe, goes with how much of the lid is seen; mobility shows no rule

Caveat contacts are counted on modelled atoms, so part of the link is built in